

	electron.
4(a)	Magnetic flux density (B) is defined as the force acting per unit current per unit length on a wire placed at right angles to the magnetic field.
4(b)(i)	Since the electronic balance registered a reduction in reading, this implies that the normal reaction force acting on it is reduced. This must be the result of the magnet experiencing an upward force due to its electromagnetic interaction with the metal rod. According to Newton's third law, if the magnet experiences an upward force, the metal rod must have experienced an equal but opposite force acting downwards. Using Fleming's left hand rule, for the given magnetic field and electromagnetic force directions, the current in the rod must be flowing from Y to X.
4(b)(ii)	The reading on the electronic balance is a measure of the normal reaction force on the balance due to the magnet. The magnitude of the normal reaction force is given by the weight of the magnet and the electromagnetic force from the magnet's interaction with the wire. When current in wire is flowing from Y to X: $W - F_B = N$ $W - F_B = (201.62 \times 10^{-3})(9.81) \text{-----}(1)$ When current in wire is flowing from X to Y: $W + F_B = N$ $W + F_B = (202.17 \times 10^{-3})(9.81) \text{-----}(2)$ $(2)-(1)$

Qn	Solution
	$2F_B = (202.17 - 201.62) \times 10^{-3} \times 9.81$ $2BIL = 0.55 \times 10^{-3} \times 9.81$ $B = \frac{0.55 \times 10^{-3} \times 9.81}{2(1.6)(0.12)}$ $B = 0.01405 \approx 0.014 \text{ T}$
4(b)(iii)	There is no change on the readings on the balance which is dependent on the amount of electromagnetic force experienced by the wire. The electromagnetic force is dependent on the magnitude of the magnetic field, length of wire perpendicular to the magnetic field, and the current flowing in the wire, which are all kept the same as before.
5(c)	Gravitational potential / at a point is the work done per unit mass in bringing a